

MAT235H-5201

Vector Calculus I

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Check Quercus for more info

Please **READ** Syllabus carefully

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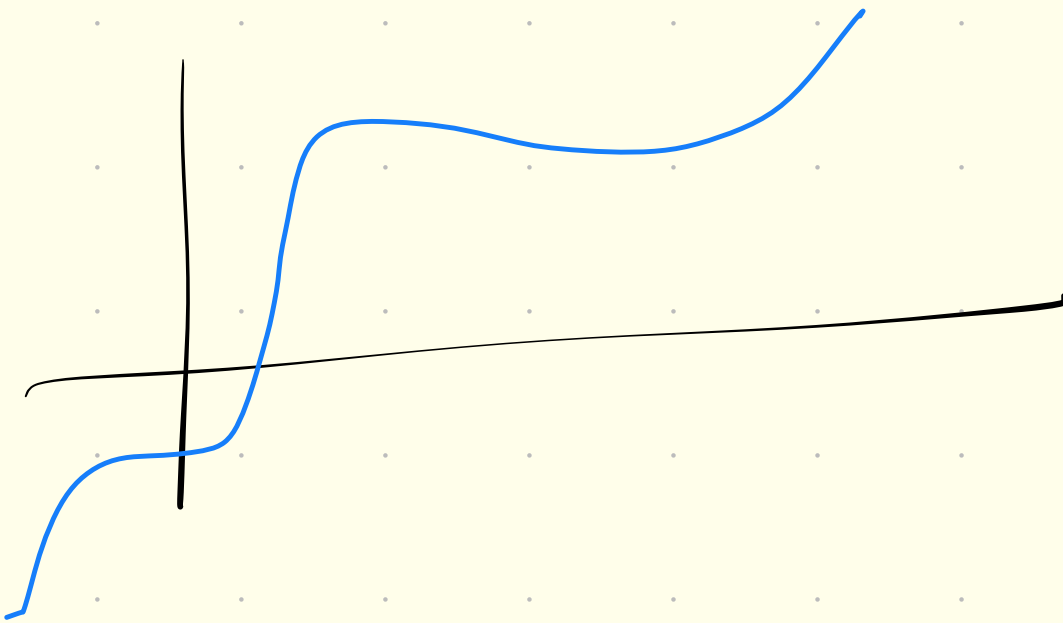
PG 101

What is multivariable calculus about?

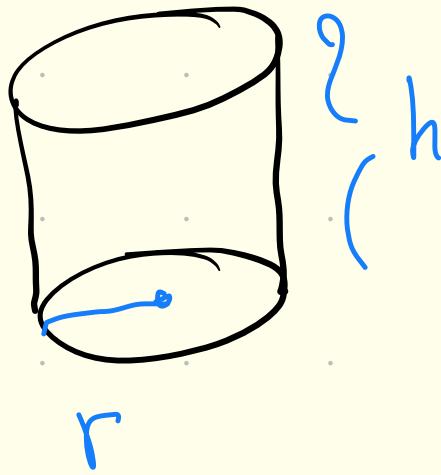
a one variable function:

Ex: polynomials; trigs, ...

x	1	2	3, 14	...
y	1	0	2	...

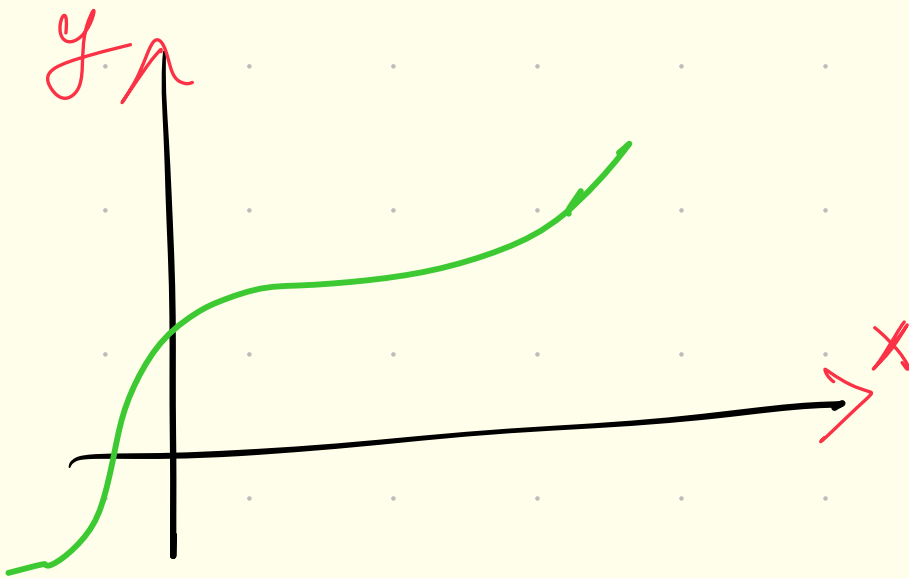


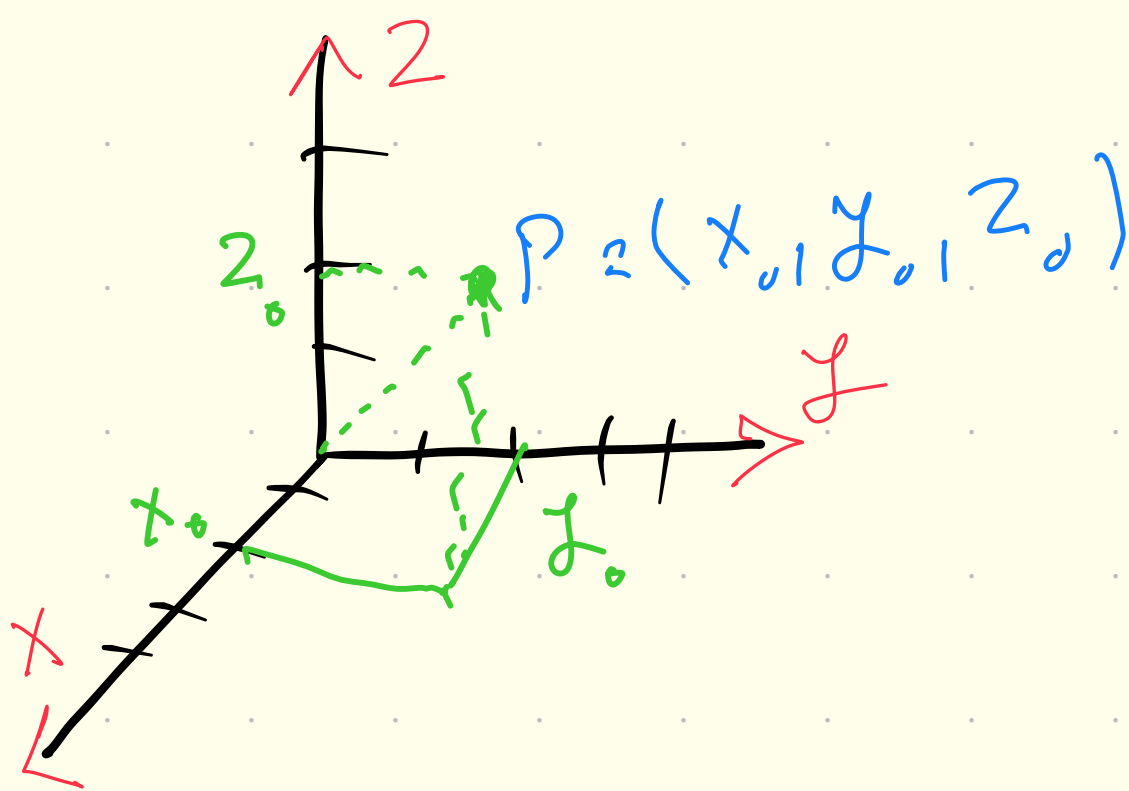
Exo



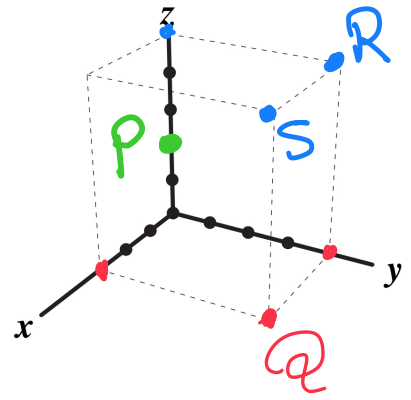
$$Vol = h \times \pi r^2$$

we need two variables

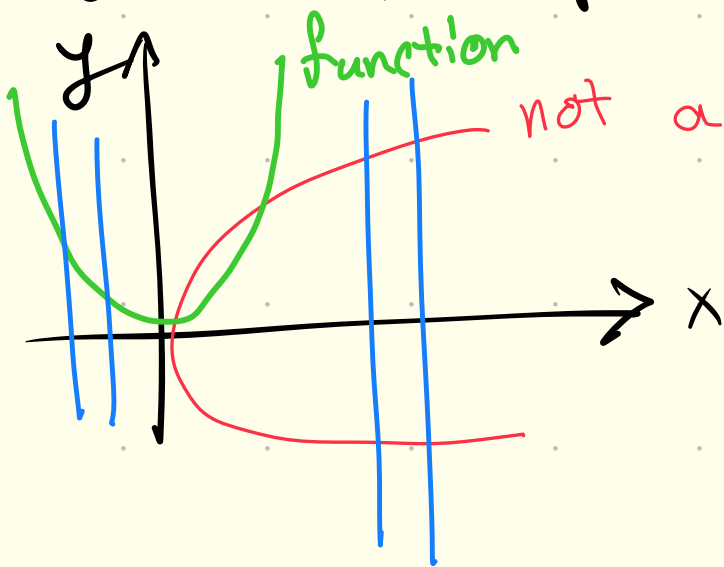




Example 1. Plot the points $P = (0, 0, 2)$, $Q = (3, 4, 0)$, $R = (0, 4, 5)$, and $S = (3, 4, 5)$ on the coordinate axes to the right.



A two variable real function is a rule to take two inputs and output one real number



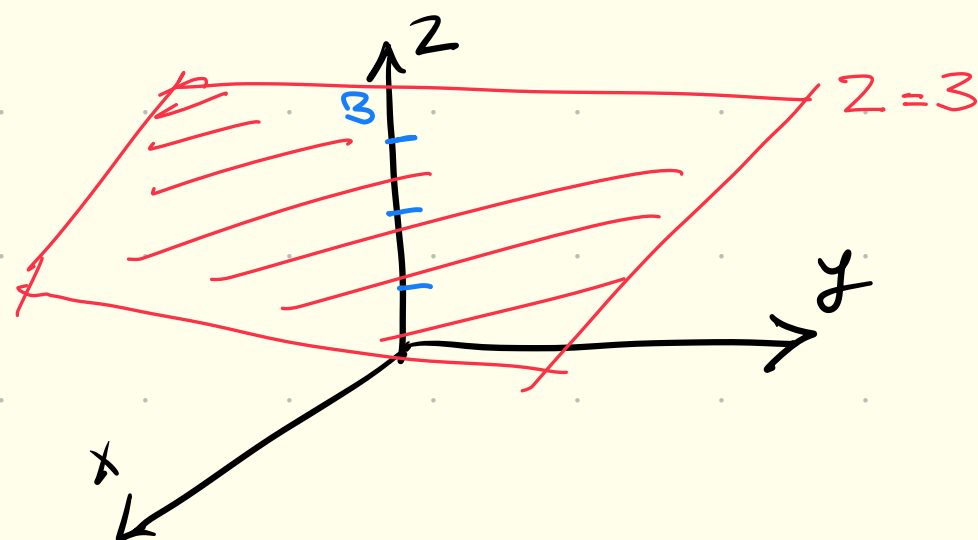
for any two inputs there is only one value

Ex: $f(x, y) = x + y$, $g(x, y) = xy, \dots$

Example 2. Consider the functions $f(x, y) = x^2 + y^2$ and $g(x, y) = 3$. Evaluate f at several values of x and y , and sketch a graph of g .

$$f(x, y) = x^2 + y^2$$

$y \backslash x$	0	1	-1
0	0	1	1
1	1	2	2
-1	1	2	2



Example 3. Windchill temperature is a temperature which tells you how cold it feels as a result of the combination of wind and temperature. Let $C = f(w, T)$, where C is the windchill temperature (in degrees Fahrenheit) that is associated with a wind speed of w miles per hour and a temperature of T degrees Fahrenheit. A table of values for the function f is given to the right:

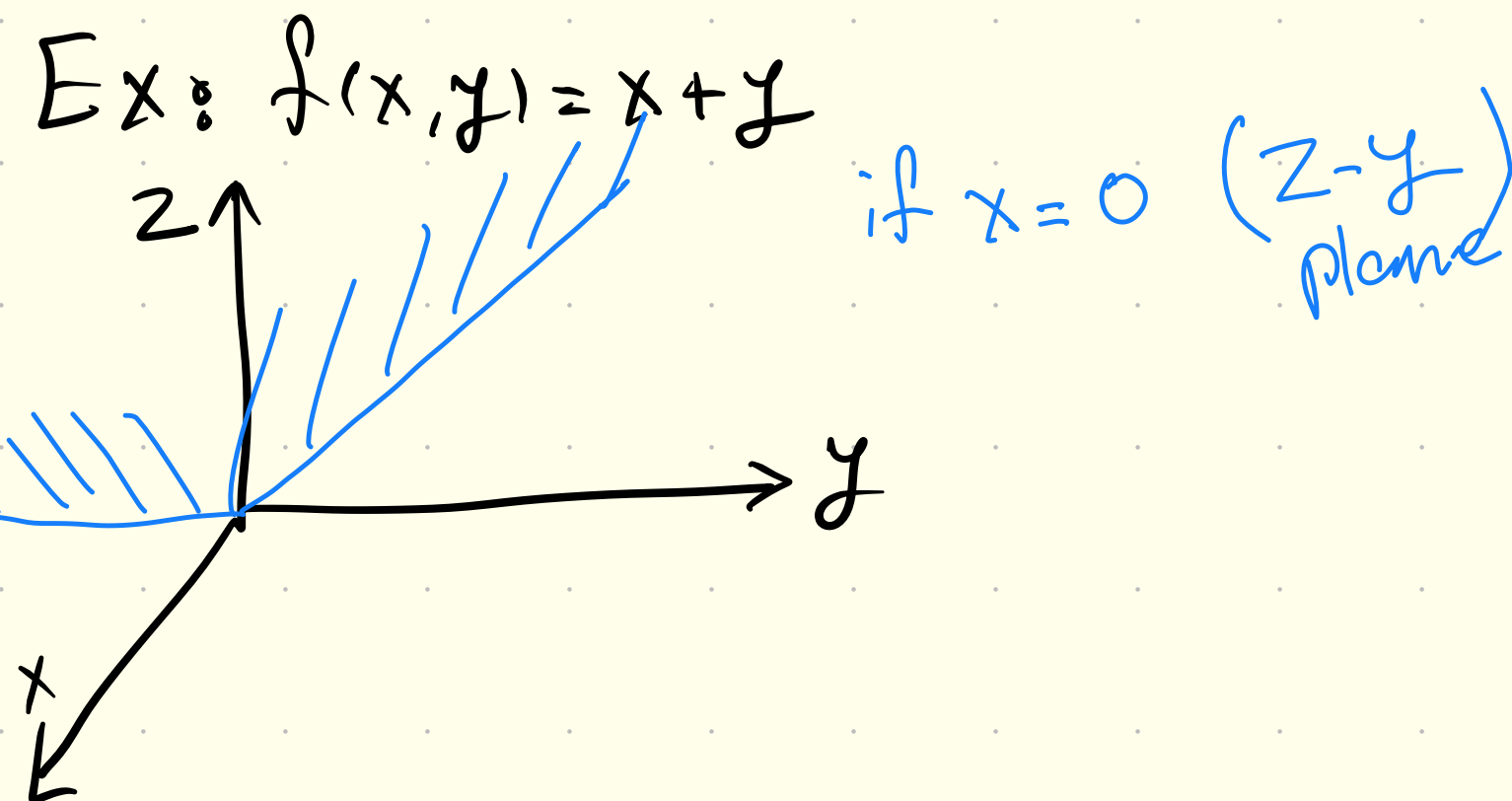
$w \setminus T$	35	30	25	20	15	10	5	0
5	31	25	19	13	7	1	-5	-11
10	27	21	15	9	3	-4	-10	-16
15	25	19	13	6	0	-7	-13	-19
20	24	17	11	4	-2	-9	-15	-22
25	23	16	9	3	-4	-11	-17	-24

- (a) Evaluate and interpret $f(20, 5)$.

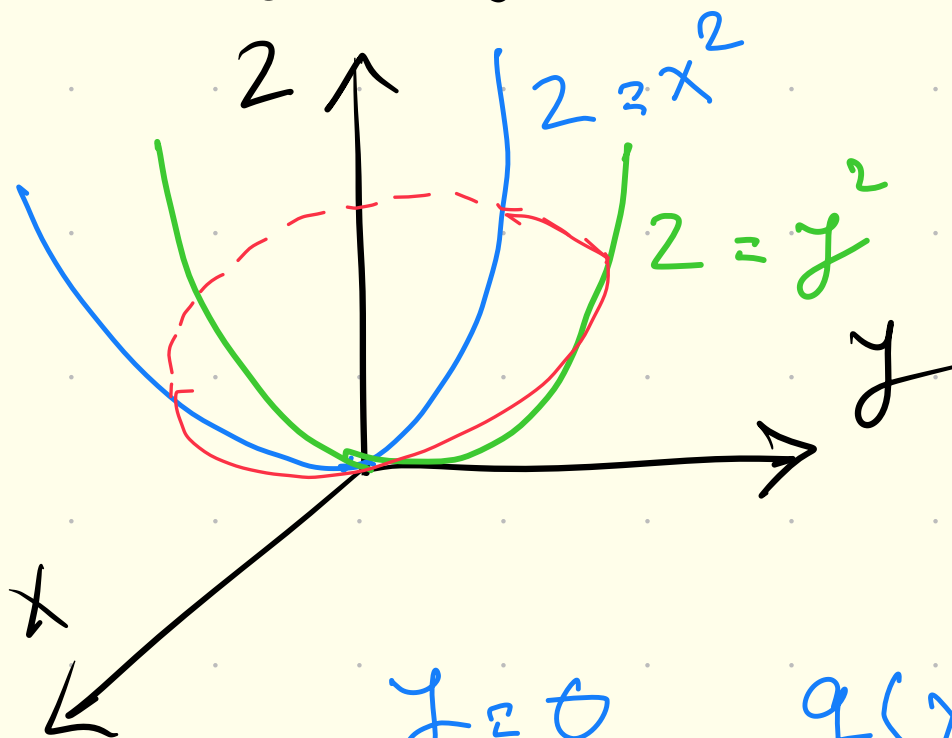
$$f(20, 5) = -15$$

- (b) How fast does the wind need to blow for it to feel like -10°F when the air temperature is really 5°F ?

$$w \approx 10 \text{ mph}$$



Ex: $g(x, y) = x^2 + y^2$



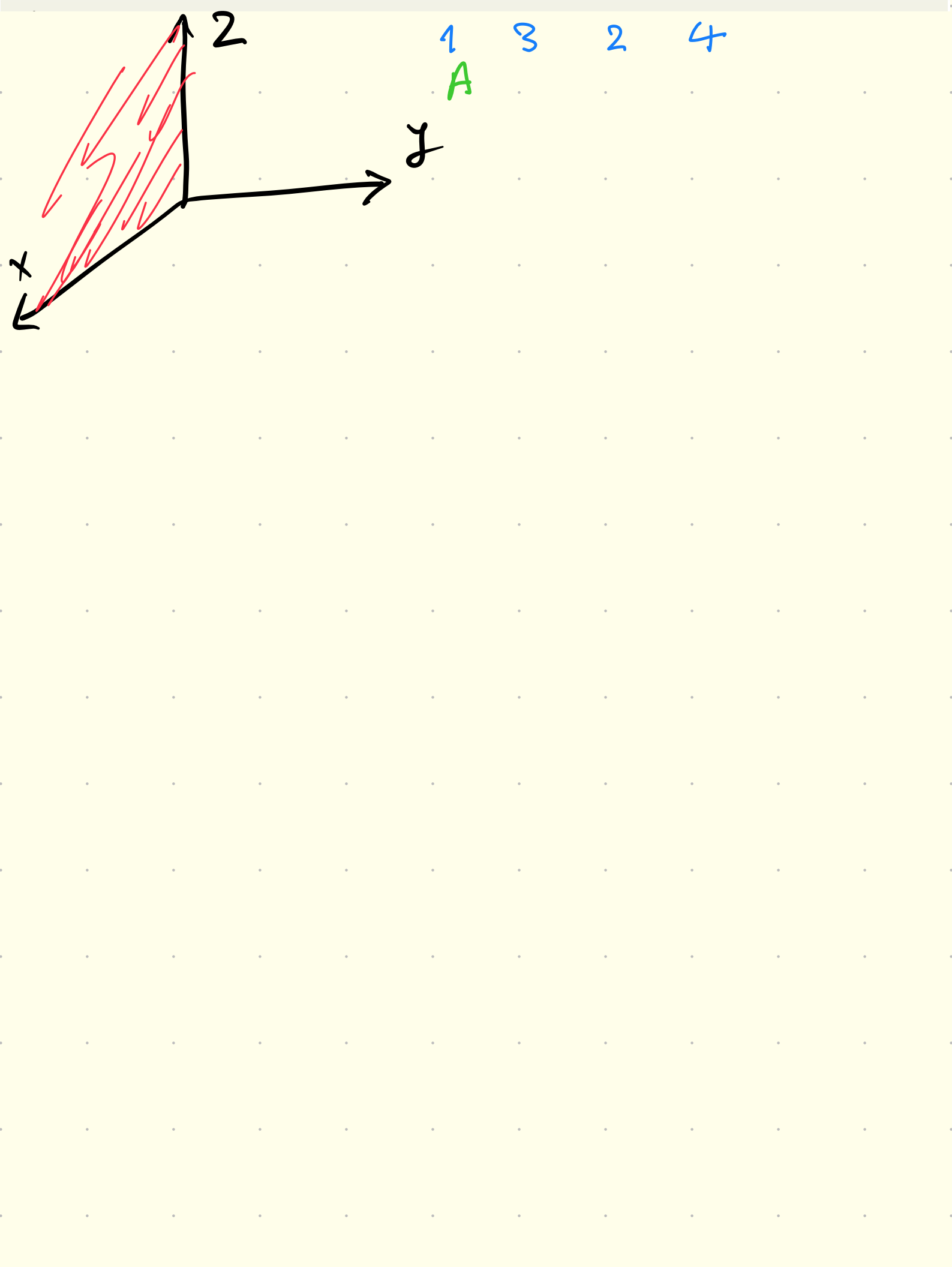
$y = 0$

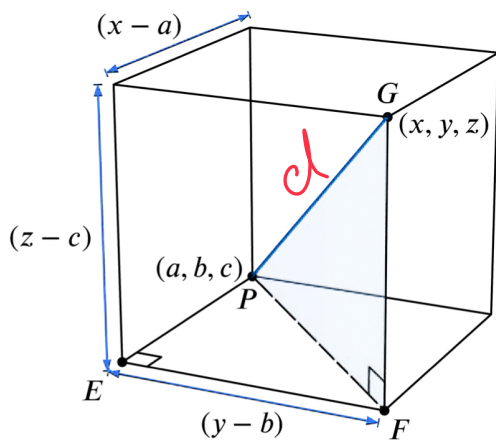
$g(x, 0) = x^2$

$x = 0$

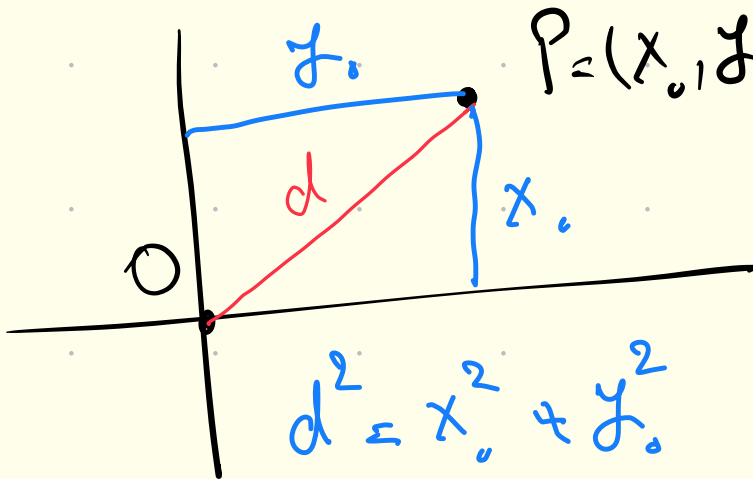
$g(0, y) = y^2$

Example 7 Which of the points $A = (1, -1, 0)$, $B = (0, 3, 4)$, $C = (2, 2, 1)$, and $D = (0, -4, 0)$ lies closest to the xz -plane? Which point lies on the y -axis?





$$d^2 = (x-a)^2 + (y-b)^2 + (z-c)^2$$



$P = (x_0, y_0)$

what is the distance of P to O?

$$d^2 = x_0^2 + y_0^2$$

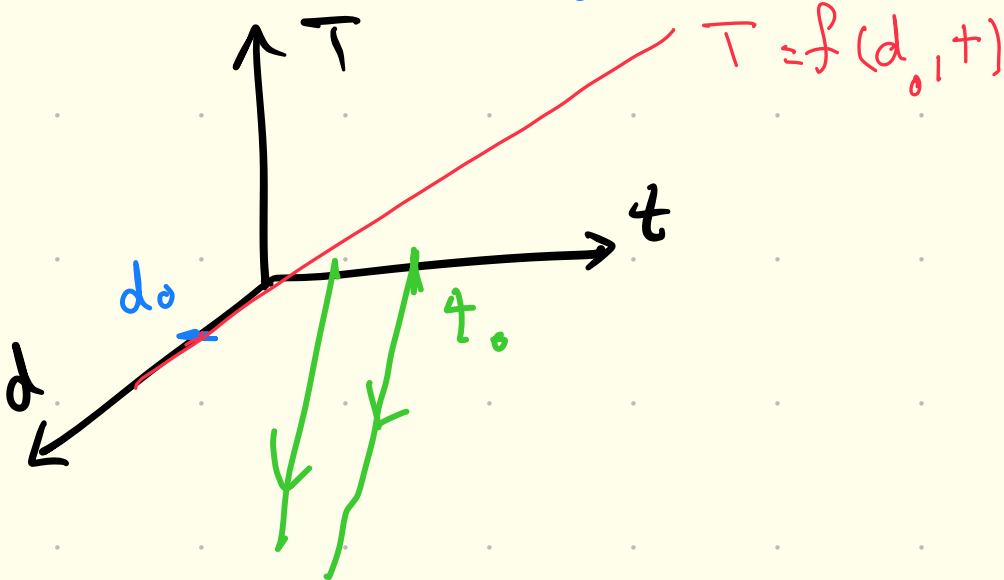
Exercises

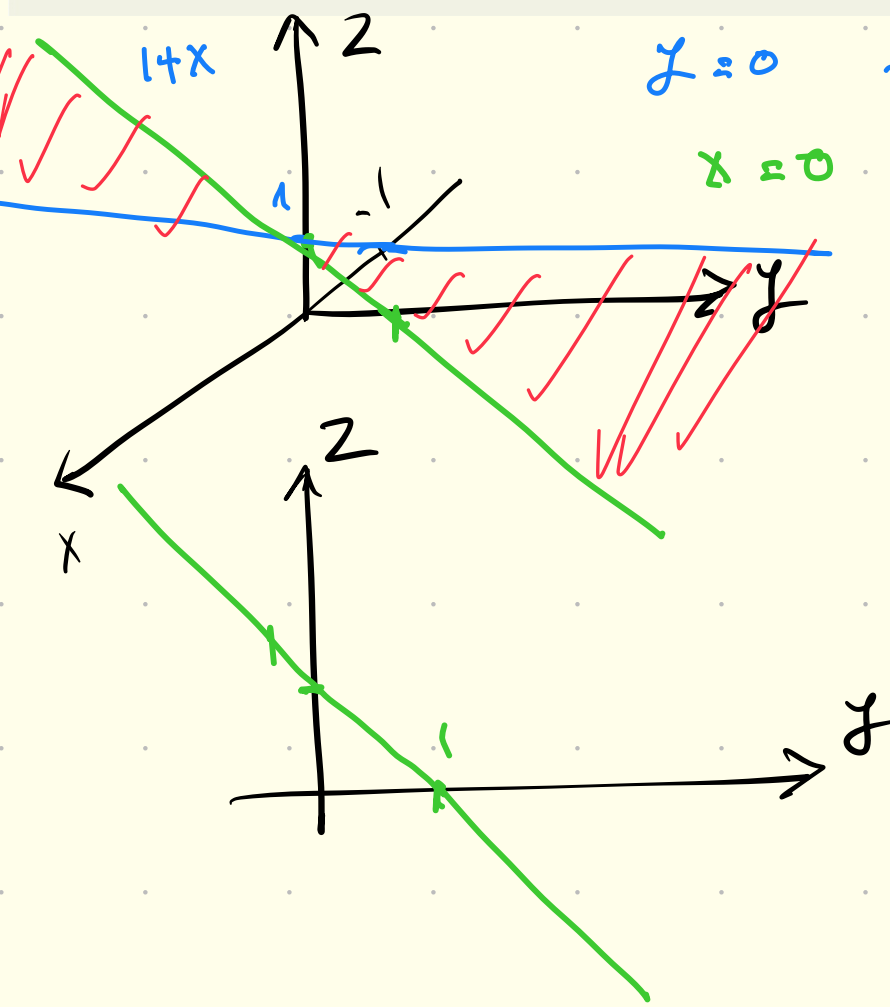
1. Let $T = f(d, t)$, where T is the temperature, in degrees Fahrenheit, of a room t minutes after a heater is turned on, at a distance of d meters from the heater.
- (a) Is T an increasing or a decreasing function of t ? Explain. Sketch possible cross sections of f for $d = 1, 2$, and 3 meters.

For a fix d , $f(d, t)$ increases $\uparrow$

- (b) Is T an increasing or a decreasing function of d ? Explain. Sketch possible cross sections of f for $t = 0, 5$, and 10 minutes.

T is decreasing function of d

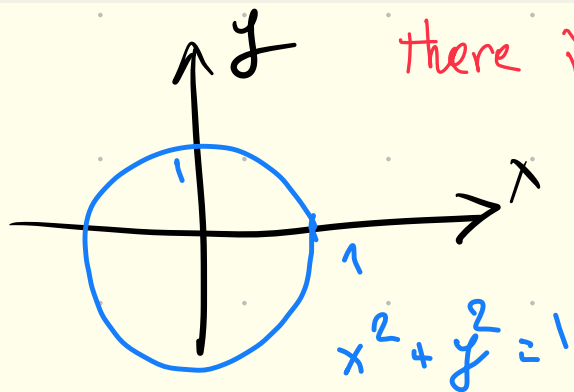


Example 4Describe the graph of $f(x, y) = 1 + x - y$.

$$z=0 \quad f(x, 0) = 1+x$$
$$x=0 \quad f(0, y) = 1-y$$

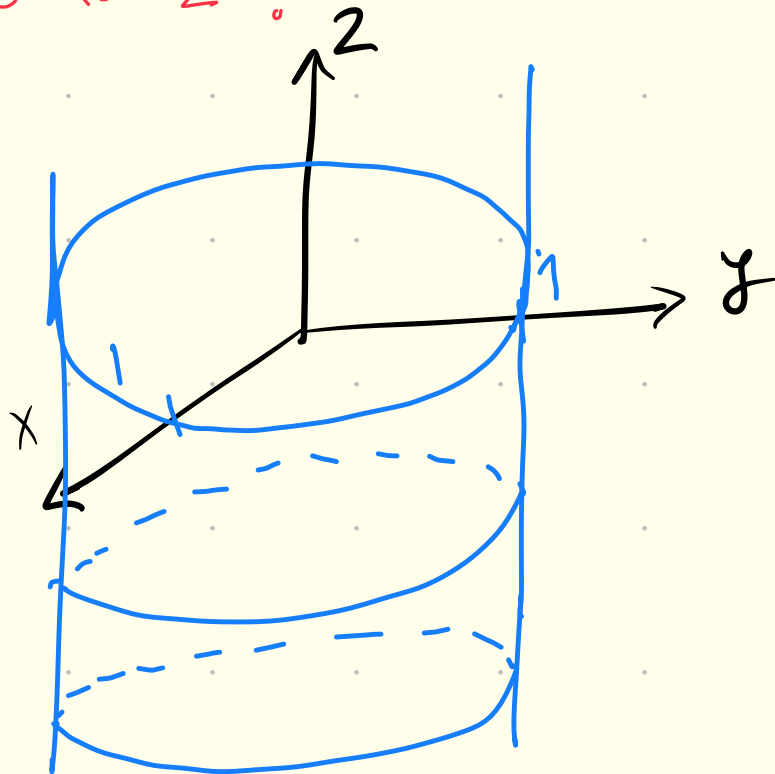
Example 5

Graph the equation $x^2 + y^2 = 1$ in 3-space.



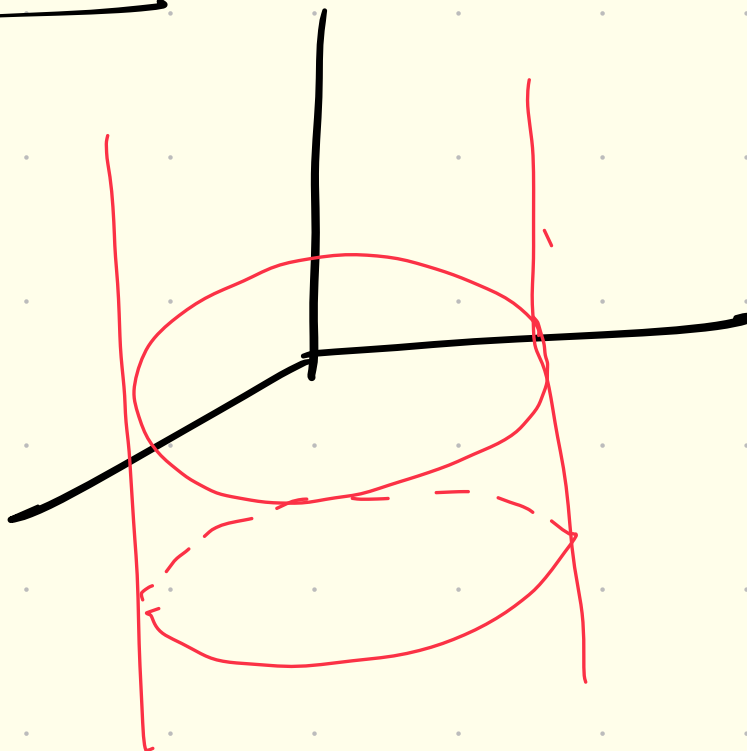
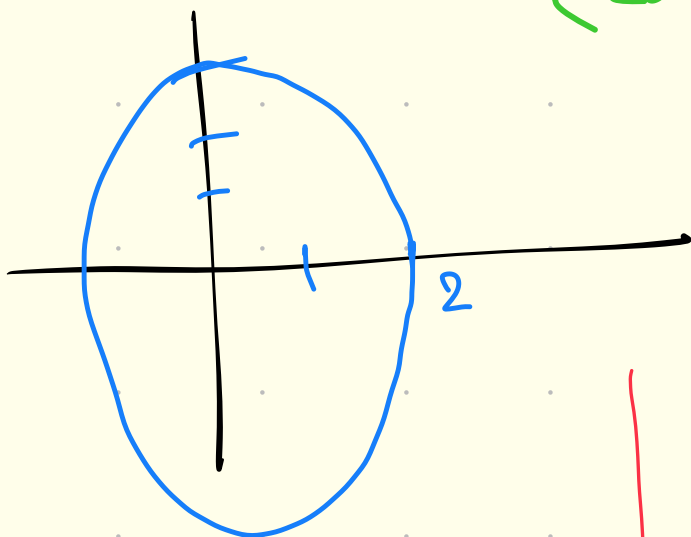
there is no z!

$$0 \cdot z = x^2 + y^2 - 1$$



If $f(x, y) = c$
(Constant)

$$\frac{x^2}{4} + \frac{y^2}{9} = 1$$



Try

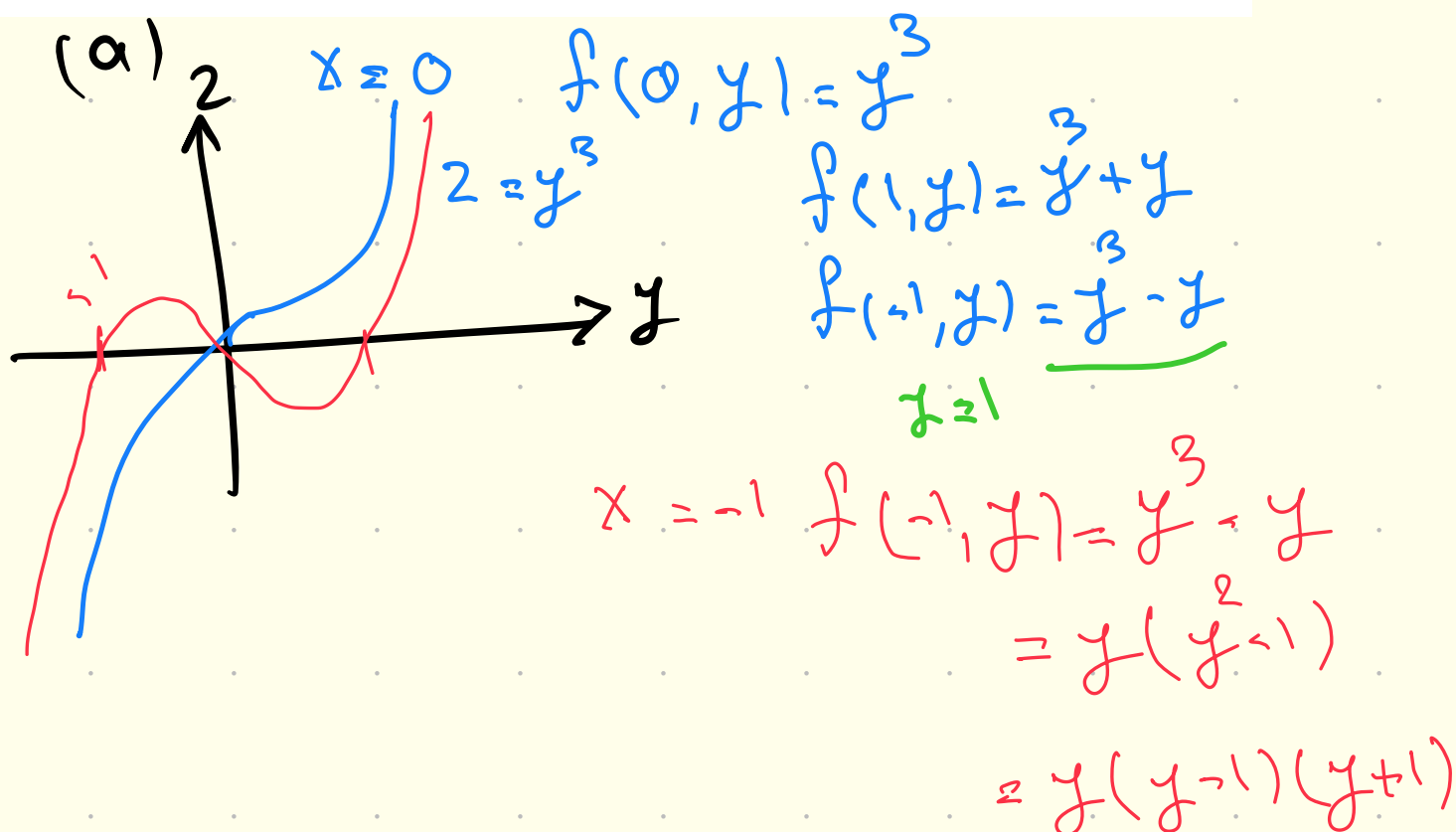
$$x^2 - y^2 = 1$$

19. Consider the function f given by $f(x, y) = y^3 + xy$.

Draw graphs of cross-sections with:

(a) x fixed at $x = -1$, $x = 0$, and $x = 1$.

(b) y fixed at $y = -1$, $y = 0$, and $y = 1$.



$$f(x, 0) = 0$$

$$f(x, 1) = 1 + x$$

$$f(x, -1) = -1 - x$$

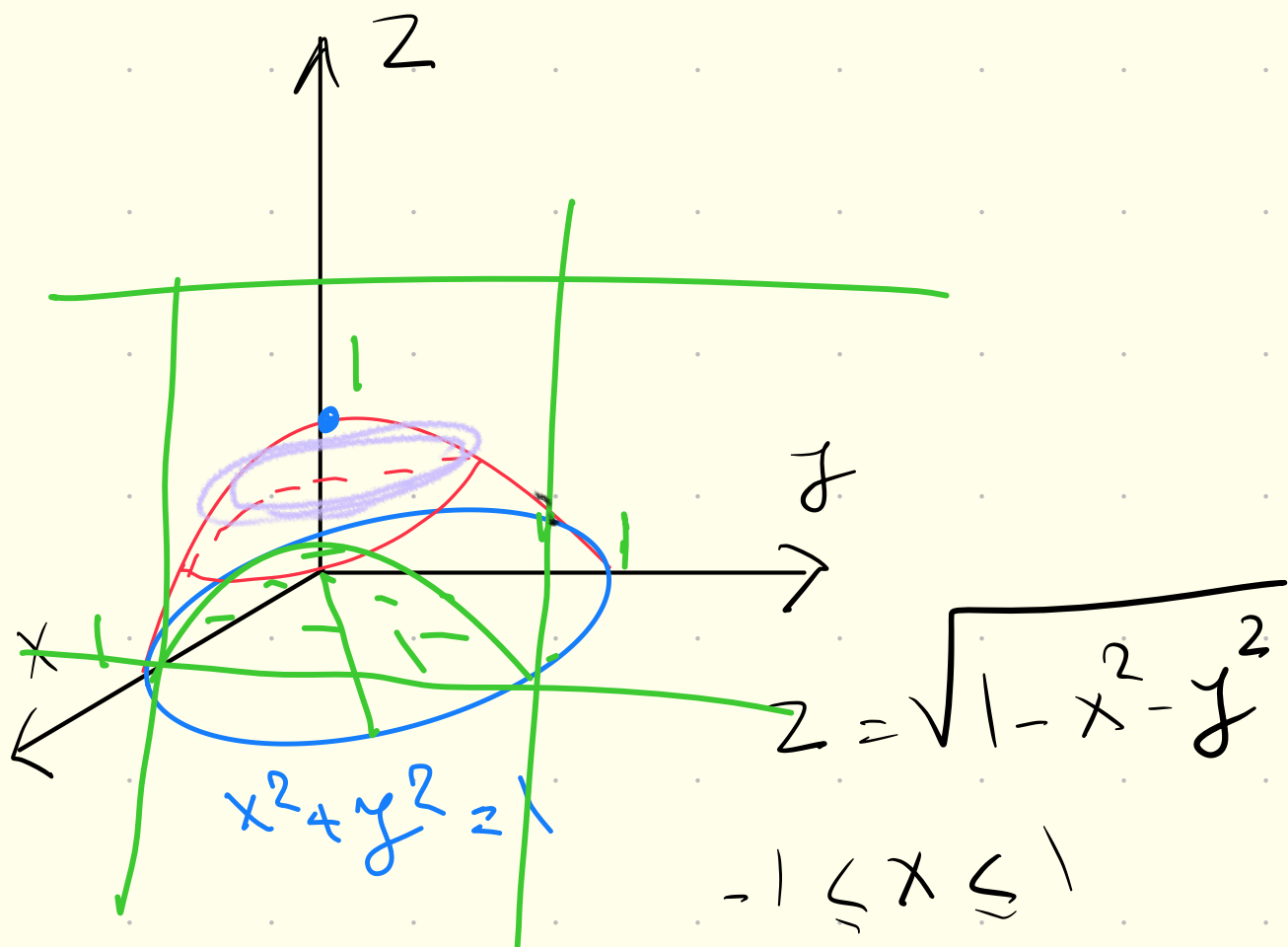
Contour:

$$z = f(x, y)$$

if $x = x_0$ constant

we get a cross-section

if $z = z_0$ we get a level curve
or contour



$$-1 \leq x \leq 1$$

$$-1 \leq y \leq 1$$

cross-section: fix x or y

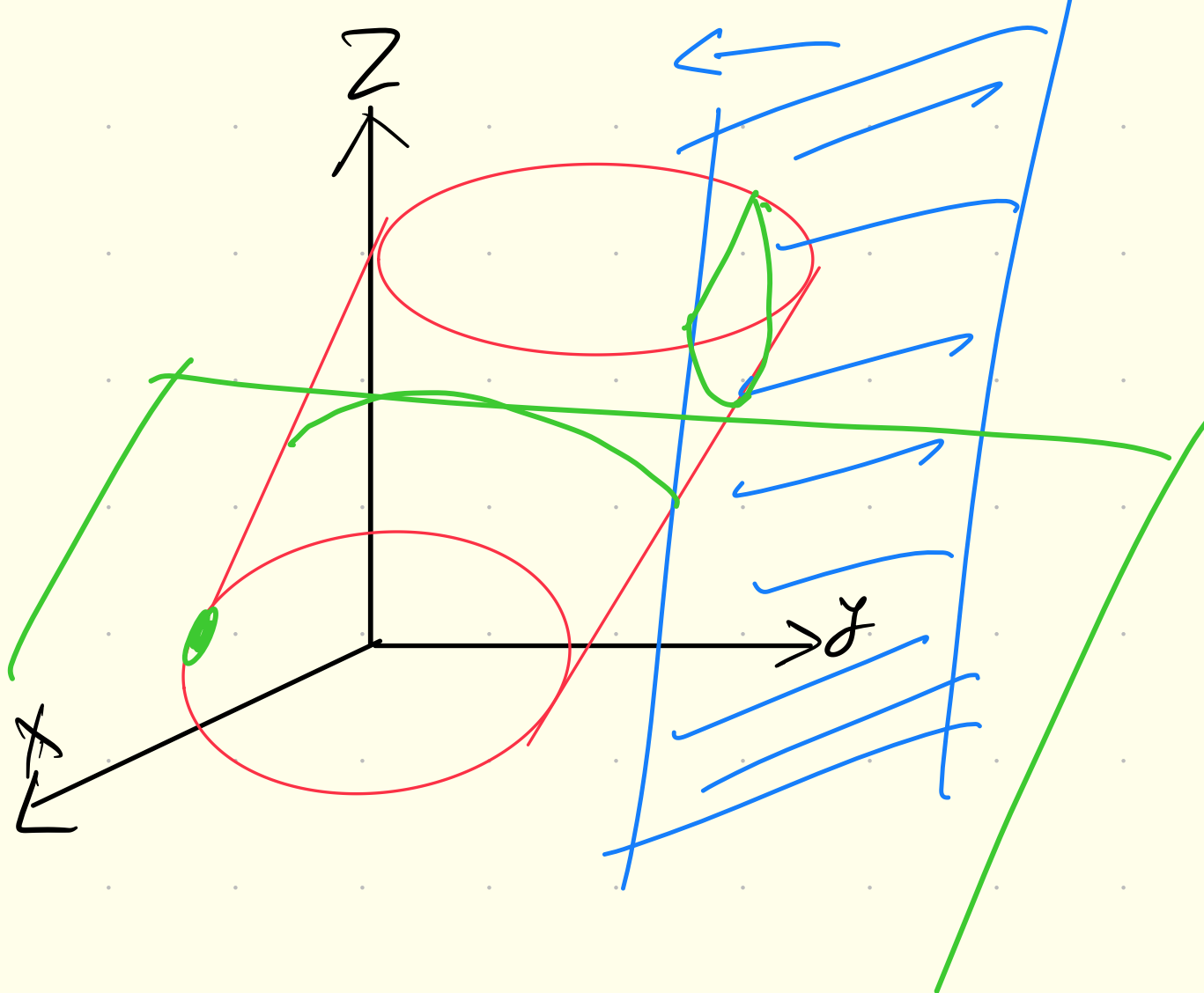
Contour: $z = \text{Constant}$

$$z = 0 \Rightarrow \sqrt{1 - x^2 - y^2} = 0 \text{ or}$$

$$x^2 + y^2 = 1 \text{ blue Curve}$$

$$z = 1 \quad \sqrt{1 - x^2 - y^2} = 1$$

$$\text{or } x^2 + y^2 = 0, \quad x = y = 0$$



$$x_0 = \frac{1}{2}$$

$$Z = \sqrt{1 - \frac{1}{4}y^2} = \sqrt{\frac{3}{4} - y^2}$$

$$Z^2 + y^2 = \frac{3}{4}$$

$$Z, Z \geq 0$$

Example 3

Find equations for the contours of $f(x, y) = x^2 + y^2$ and draw a contour diagram for f . Relate the contour diagram to the graph of f .

$$z = x^2 + y^2 \geq 0$$

$$z = 0 \Rightarrow x = y = 0$$

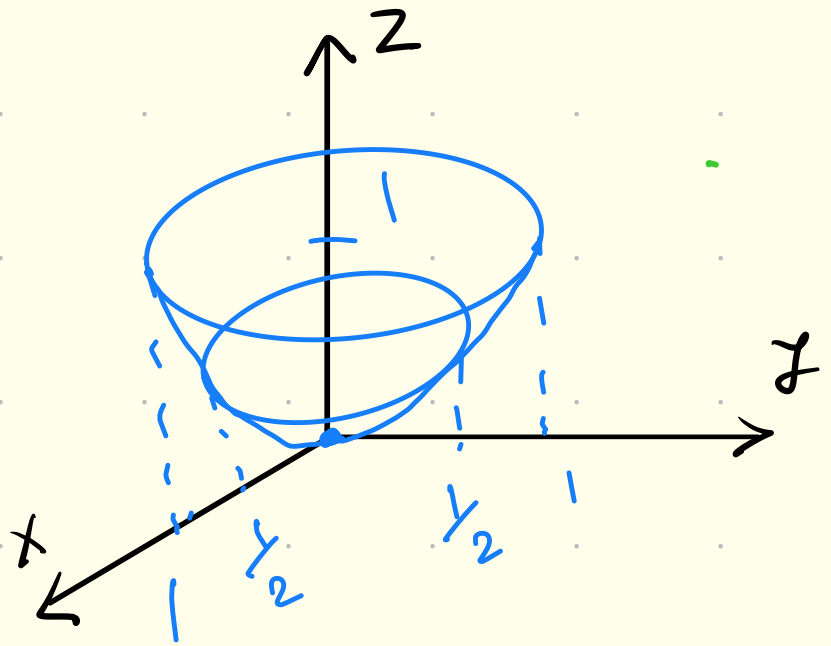
$$z = 1 \Rightarrow x^2 + y^2 = 1$$

$$x^2 + y^2 = r^2$$

circle of radius r

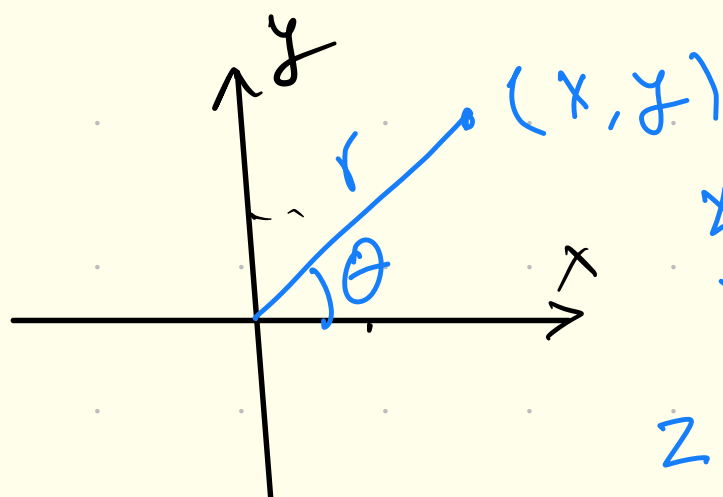
$$z = \frac{1}{4}$$

$$x^2 + y^2 = \frac{1}{4}$$

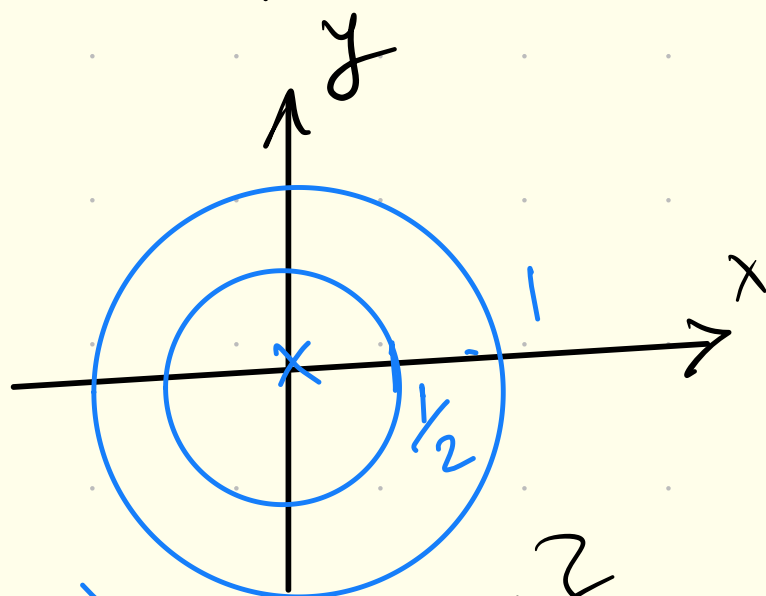


Example 4

Draw a contour diagram for $f(x, y) = \sqrt{x^2 + y^2}$ and relate it to the graph of f .



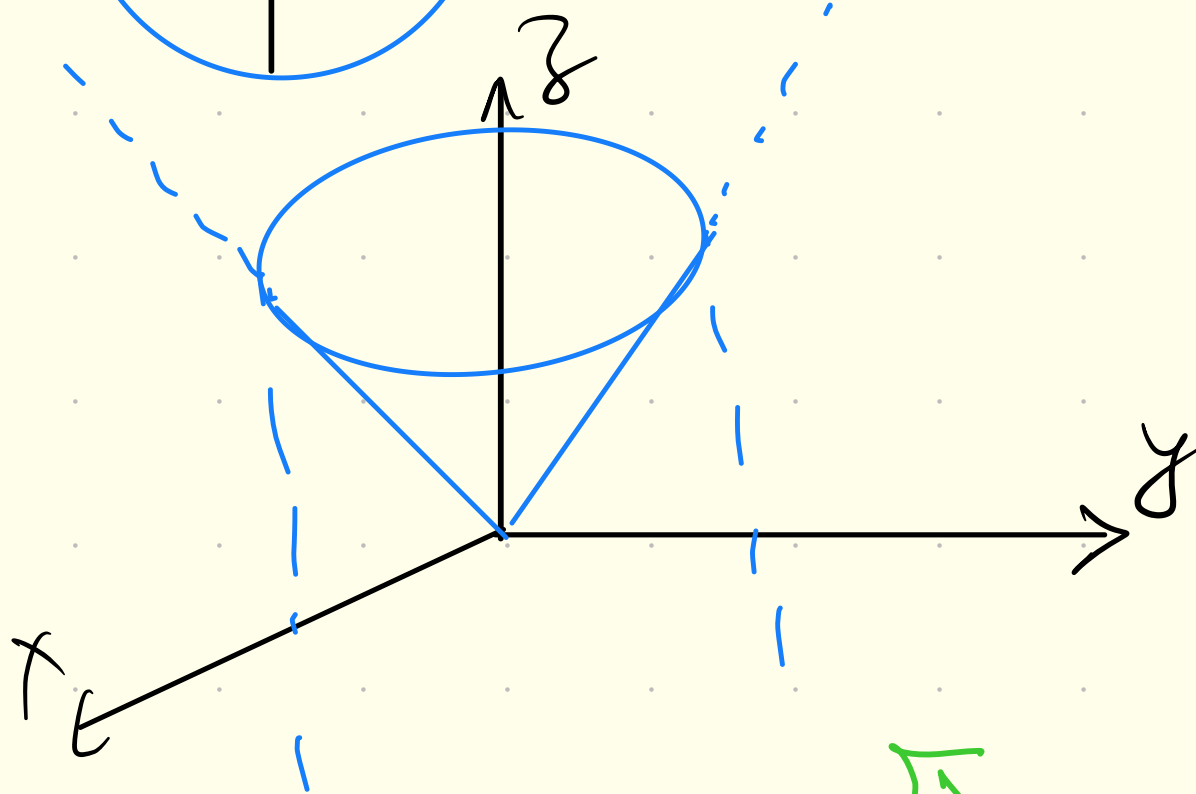
$$x = r \cos \theta$$
$$y = r \sin \theta$$
$$z = r$$



$$z = 0 \Rightarrow x = y = 0$$

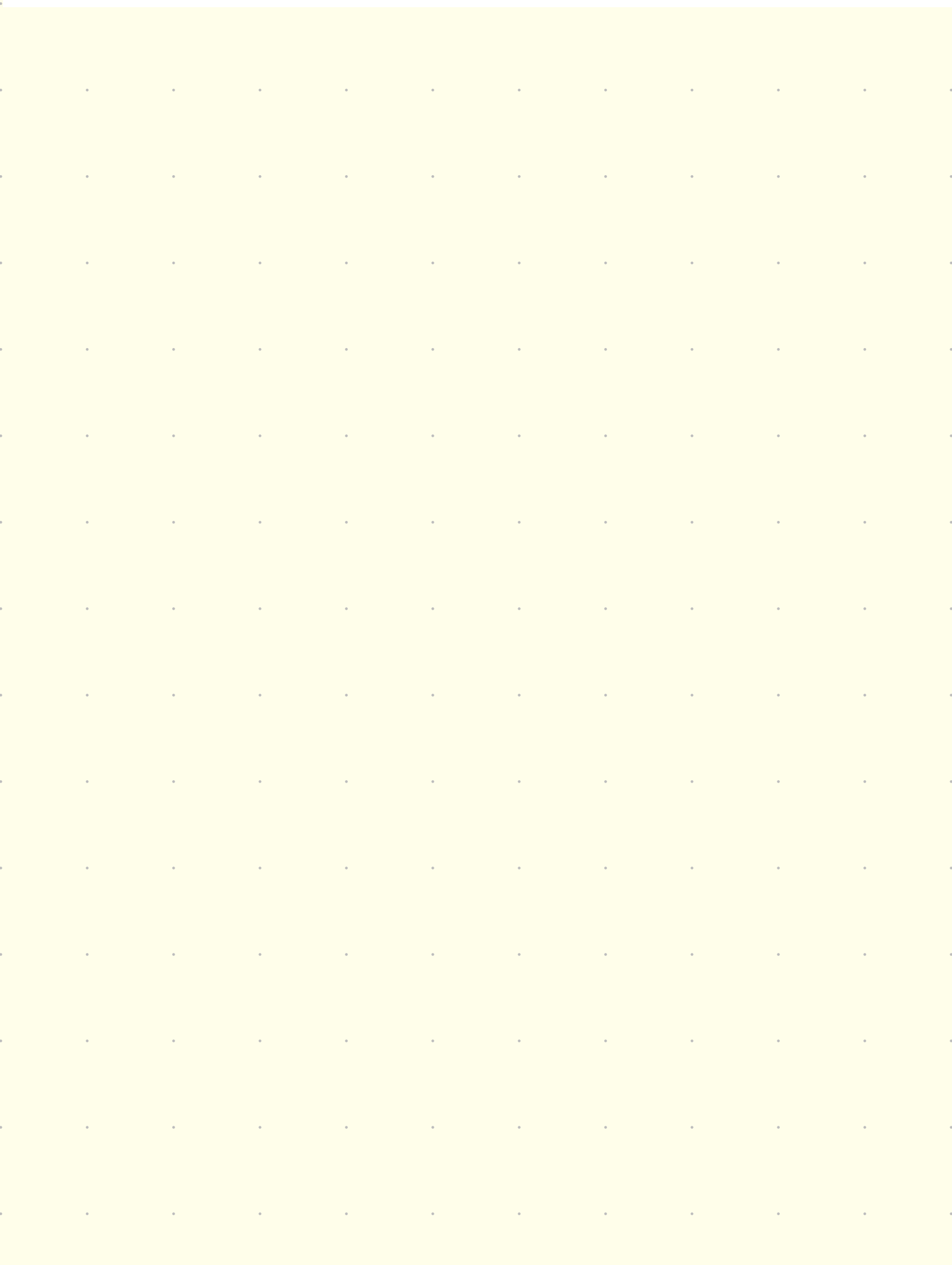
$$z = 1 \Rightarrow x^2 + y^2 = 1$$

$$z = 1/2 \Rightarrow x^2 + y^2 = 1/4$$



$$x = 0 \Rightarrow z = |y|$$

2. Make a contour plot for the function $z = 2y - x$ with at least 3 different contours. Label each contour with the corresponding value of z .



Example 6 Relate the values of $f(x, y) = x^2 - y^2$ in Table 12.5 to its contour diagram in Figure 12.48.

Table 12.5 Table of values of $f(x, y) = x^2 - y^2$

y	3	0	-5	-8	-9	-8	-5	0
	2	5	0	-3	-4	-3	0	5
	1	8	3	0	-1	0	3	8
	0	9	4	1	0	1	4	9
	-1	8	3	0	-1	0	3	8
	-2	5	0	-3	-4	-3	0	5
	-3	0	-5	-8	-9	-8	-5	0
		-3	-2	-1	0	1	2	3
x								

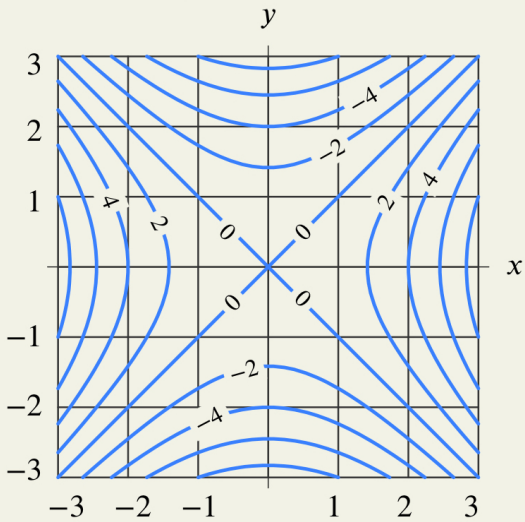


Figure 12.48: Contour map of $f(x, y) = x^2 - y^2$

Example 1. An airline sells discount fares and regular fares. Let d be the number of discount fares the airline sells, let f be the number of regular (full) fares the airline sells, and let R be the resulting revenue, in thousands of dollars. The table to the right gives R as a function of d and f . Find a formula for R .

		Regular fares, f			
		0	100	200	300
Discount fares, d	0	0	24	48	72
	100	8	32	56	80
	200	16	40	64	88
	300	24	48	72	96

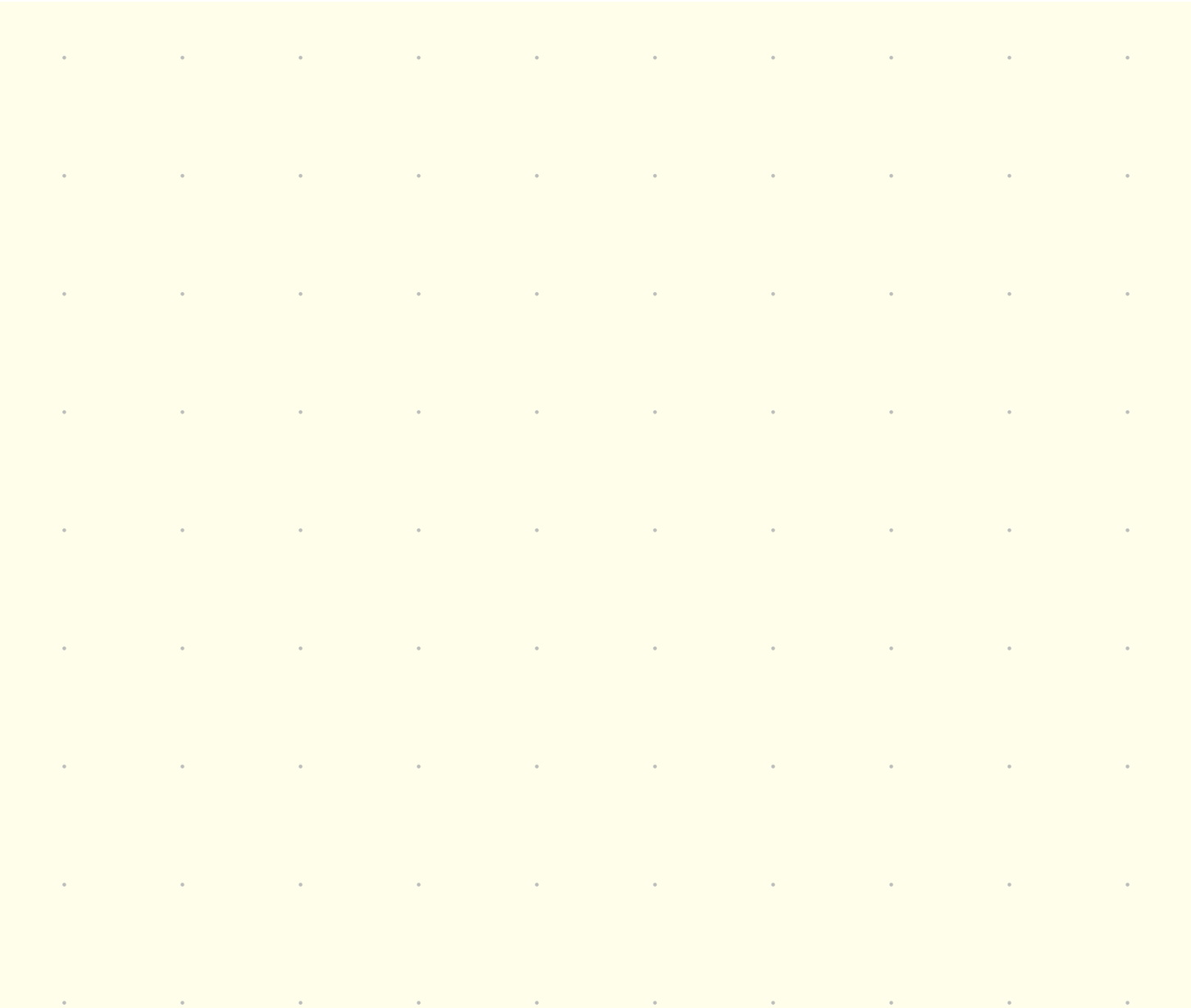
$$\text{Slope in "f" direction} = \frac{\Delta R}{\Delta f} = \frac{24 - 0}{100 - 0} = 0.24 \leftarrow \text{same for every row}$$

$$\text{Slope in "d" direction} = \frac{\Delta R}{\Delta d} = \frac{8 - 0}{100 - 0} = 0.08 \leftarrow \text{same for every column}$$

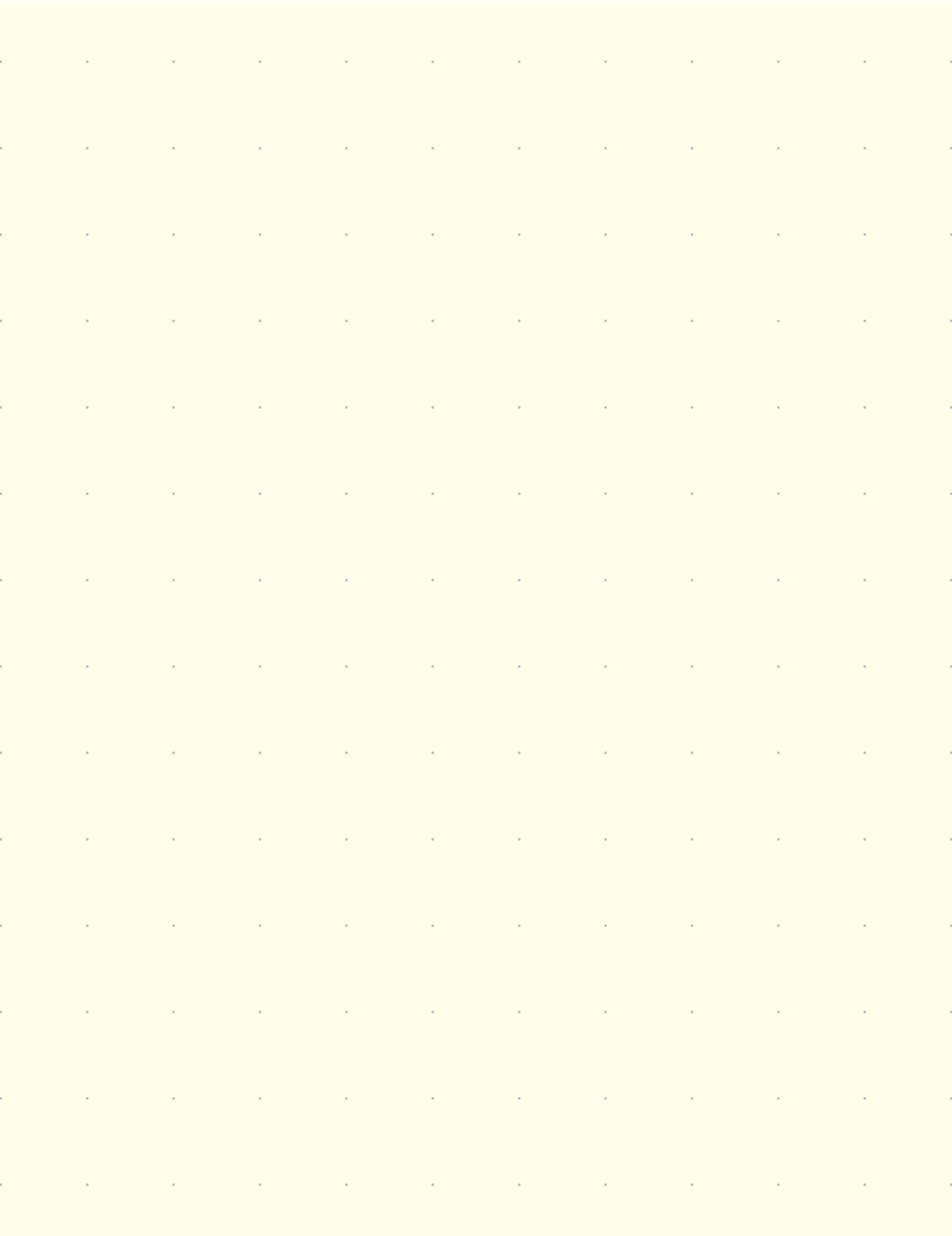
Since every row is linear with the same slope and every column is linear with the same slope, we know that $R = g(d, f)$ is a linear function. Therefore, by the calculations above, we have $m = 0.08$ and $n = 0.24$, and we observe from the table that $(d_0, f_0, R_0) = (0, 0, 0)$ is a point on the graph of this function. Therefore, by the preceding formula, we have

$$R = 0 + 0.08(d - 0) + 0.24(f - 0) = 0.08d + 0.24f,$$

so $R = 0.08d + 0.24f$ gives a formula for the function g .



Example 2. Find the equation of the plane that goes through the points $(4, 0, 0)$, $(0, 3, 0)$, and $(0, 0, 2)$.

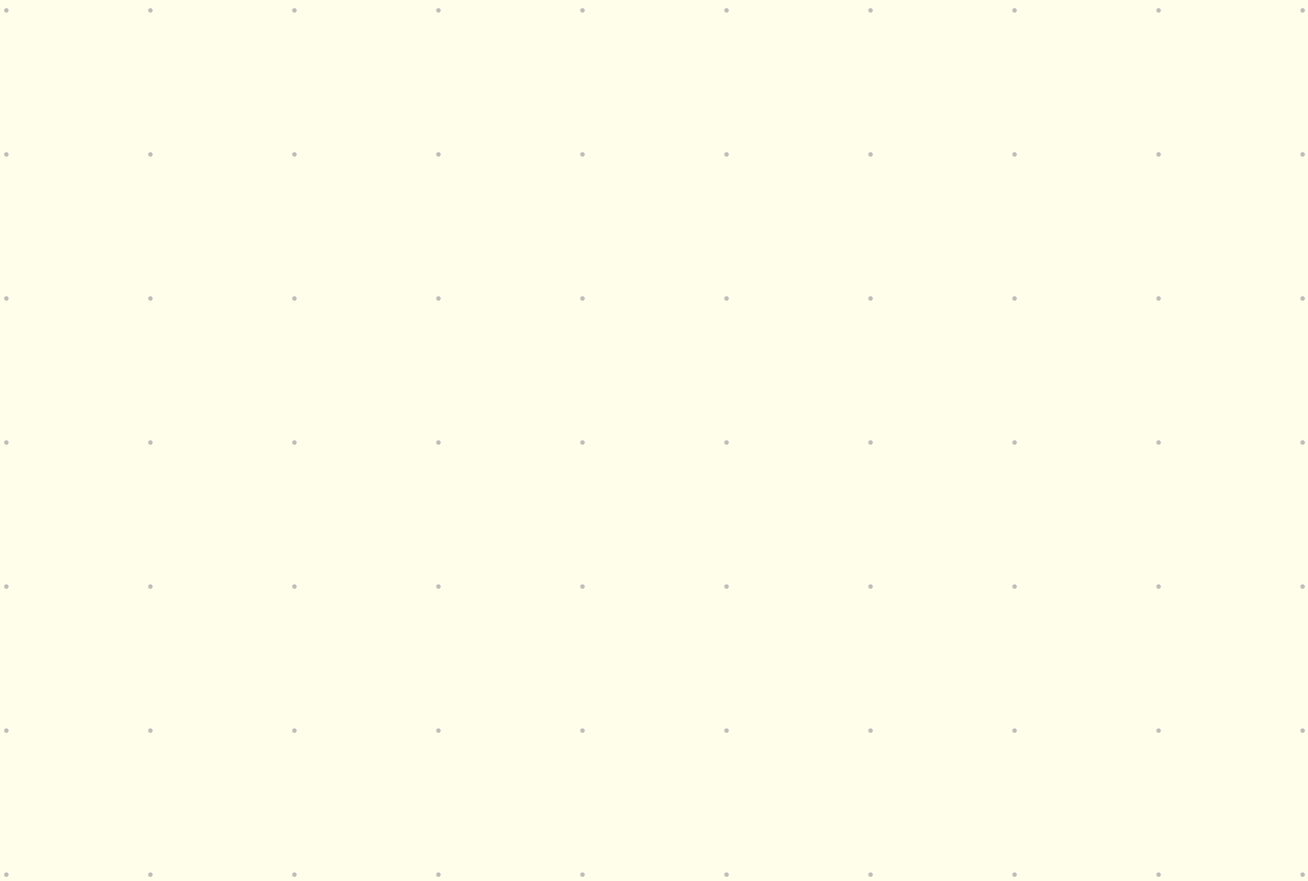


Example 3. Which of the tables of values below could represent linear functions? Explain.

$x \setminus y$	0	5	10
1	2	4	6
5	4	8	12
9	8	16	32

$x \setminus y$	0	3	6
0	1	-4	-9
2	4	-1	-6
4	7	2	-3
6	10	5	0

$x \setminus y$	0	1	2	3
0	1	1	1	1
1	2	2	2	2
2	1	1	1	1
3	2	2	2	2



Preliminary Example B. Sketch graphs of the following three surfaces in three-dimensional space:
 $x^2 + y^2 = 4$, $x^2 + z^2 = 9$, and $y = x^2$.

